**8** The pre-release material contains information on Pulse Rate and Body Mass Index (BMI). A student is investigating whether there is a relationship between pulse rate and BMI. A **section** of the available data is shown in the table.

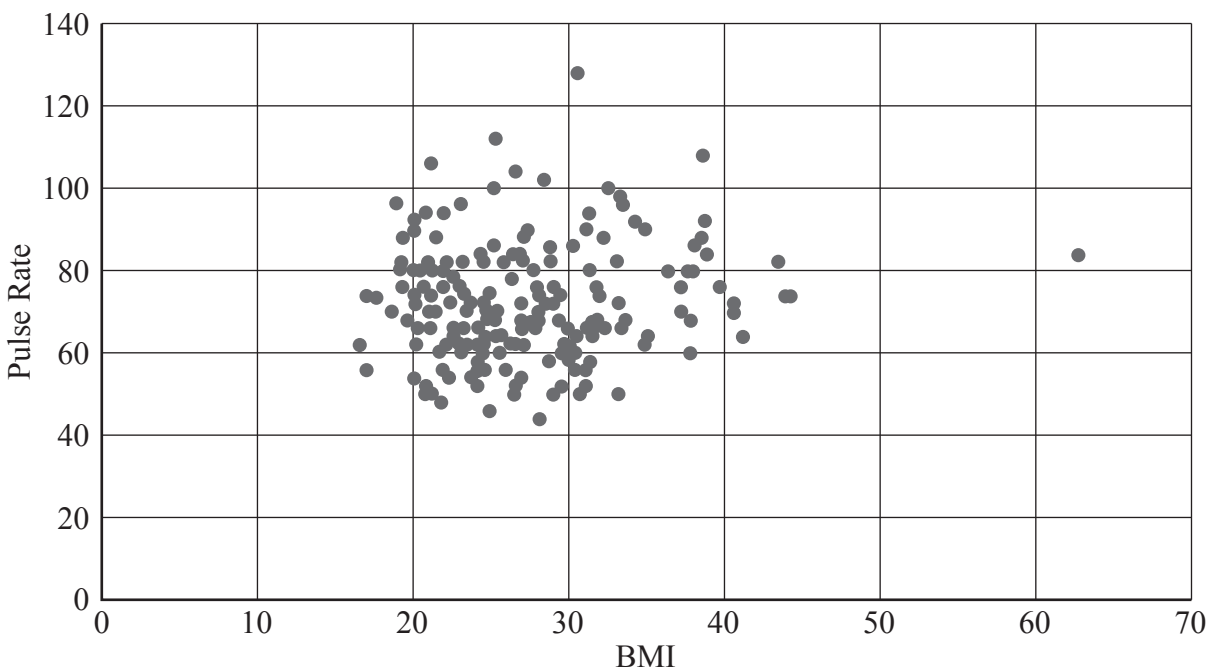
| Sex | Age | BMI | Pulse |
|---|---|---|---|
| Male | 62 | 29.54 | 60 |
| Female | 20 | 23.68 | #N/A |
| Male | 17 | 26.97 | 72 |
| Male | 35 | 24.7 | 64 |
| Male | 17 | 20.09 | 54 |
| Male | 85 | 23.86 | 54 |
| Female | 81 | 24.04 | #N/A |

The student decides to draw a scatter diagram.

**(a)** With reference to the table, explain which data should be cleaned before any analysis takes place. **[1]**

The student cleans the data for BMI and Pulse Rate in the pre-release material and draws a scatter diagram.

Scatter diagram of Pulse Rate against BMI




The student identifies **one** outlier.

**(b)** On the copy of the scatter diagram in the Printed Answer Booklet, circle this outlier. **[1]**

The student decides to remove this outlier from the data. They then use the LINEST function in the spreadsheet to obtain the following formula for the line of best fit.

$P = 0.29Q + 64.2,$

where $P =$ Pulse Rate and $Q =$ BMI.

They use this to estimate the Pulse Rate of a person with BMI 23.68.
They obtain a value of 71 correct to the nearest whole number.

**(c)** With reference to the scatter diagram, explain whether it is appropriate to use the formula for the line of best fit. **[1]**

It is suggested that all pairs of values where the pulse rate is above 100 should also be cleaned from the data, as they must be incorrect.

**(d)** Use your knowledge of the pre-release material to explain whether or not all pairs of values with a pulse rate of more than 100 should be cleaned from the data. **[1]**